

## Lab 3: Architectural Justification

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**ArchitectureSelection:** We chose **Layered Architecture** for the Coffee Kiosk System.

### Architectural Choice

Three-Layer Architecture, which consists of the following three layers:

Presentation Layer (User Interface component)

Business Layer (Order Manager, Payment service and Receipt printer components)

Data Layer (Database Component).

All components communicate exclusively via provided and required interfaces; therefore, each component is replaceable without affecting the others.

### Two Reasons

- **The kiosk maps naturally onto layers.** There will be a touch screen interface, ordering logic (3 coffees, 2 sizes, and card only payment), receipt hardware, and stored data for menus and prices in the scenario. All of these are mapped to different layers making it easy to design, develop and understand the system. Using microservices for five smaller pieces is a lot to do.
- **It runs on a single device.** There is only one machine known as the kiosk; hence nothing will need to be scaled or implemented separately. The client-server approach will introduce an extra server component, which will become a single point of failure because once the network fails, the kiosk will not accept orders.

### Security Advantage

The Order Manager is the sole point of entry for the layers to communicate with each other. The touch screen will never have access to the database or payment service in any way; all of the requests have to be verified in one single point. Card processing happens exclusively in the Payment Service module, and everything else receives either approve or decline notifications.

### Performance Benefit

All components run on the same machine; therefore, there are no network latencies involved in communications between them. Menu and prices information is small and read-mostly, so access to it is very fast and can be kept in memory. It ensures the responsiveness of the touch screen in busy café hours.